

In-Store Customer Movement

& Product Placement Intelligence

Using H&M transaction data to map how customers move through a store and recommend exactly where every product should be placed — backed by a predicted revenue number.

31M Transactions	1.37M Customers	105K Articles	~35 GB Dataset Size	Real H&M; Data
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1 Theme

We treat 31 million H&M purchase records as a physical movement map. The sequence in which a customer buys items reflects the path they walked through the store. We load this into a graph database, run graph algorithms to reveal movement patterns, and produce a ranked list of exactly which products should move where — with a predicted revenue number for every single recommendation.

The Core Insight	Graph Algorithms Used	
Purchase sequence = movement path. Buy tops then jeans Co-purchase = adjacency signal. If 70% of customers buy Low-traffic category = dead zone. Products placed there a High PageRank product = gateway item. Triggers a whole	Degree Centrality	Finds dead zones — areas no customer reaches
	Betweenness Centrality	Finds corridor zones — perfect for impulse buys
	Louvain Detection	Groups zones always visited together
	PageRank on Products	Finds gateway products that trigger large baskets

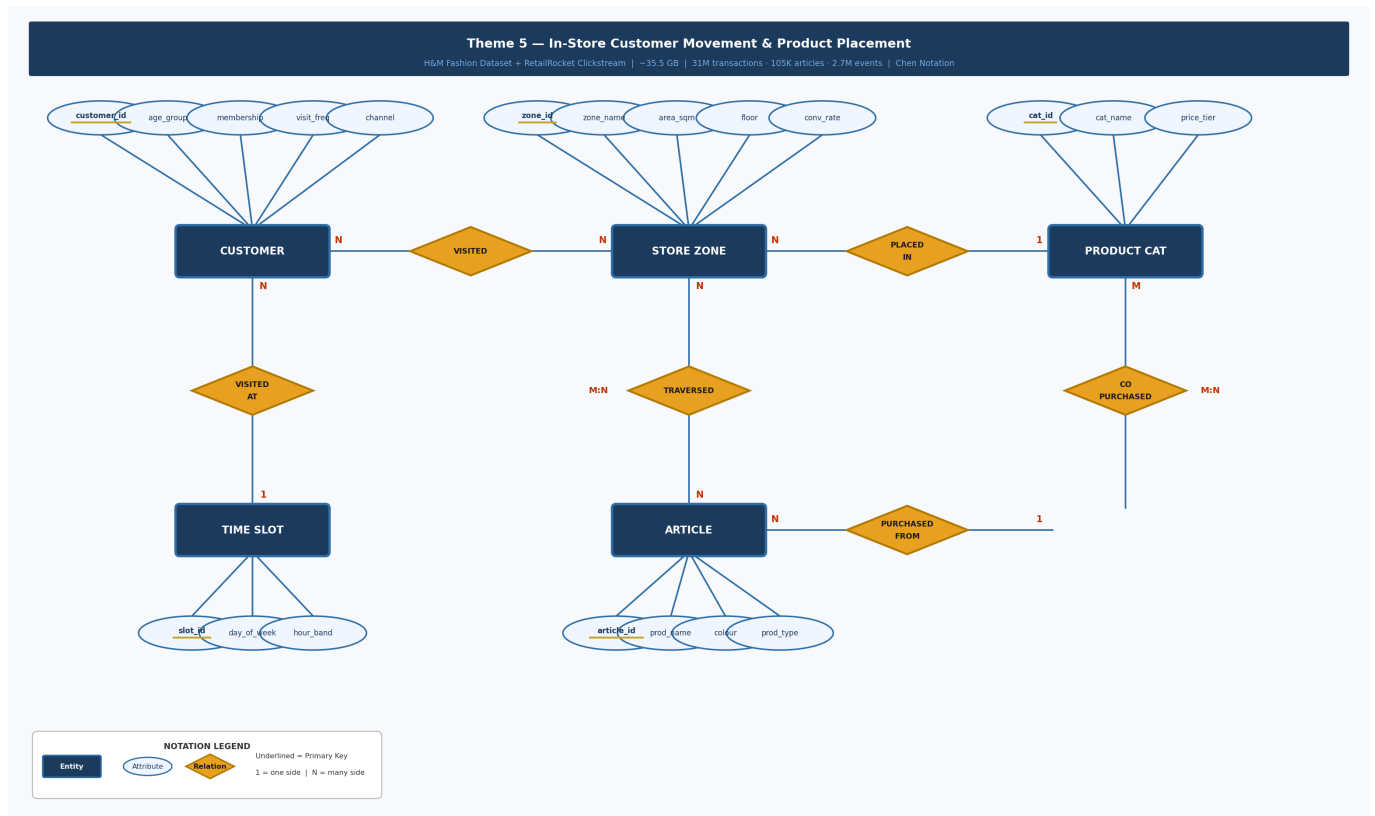
2 Dataset

	H&M; Fashion Dataset	RetailRocket Clickstream
Source	Kaggle — H&M; Personalised Fashion Recommendations	Kaggle — RetailRocket E-Commerce Dataset
Link	kaggle.com/competitions/h-and-m-personalized-fashion-recommendations	kaggle.com/datasets/retailrocket/e-commerce-dataset
Size	~35 GB (transactions + images + metadata)	~500 MB (2.75M clickstream events)
Records	31M transactions 1.37M customers 105K articles	2.75M events 1.4M visitors 235K products

	H&M; Fashion Dataset	RetailRocket Clickstream
What it gives	Real H&M; purchase sequences — proxy for physical movement	View → Add-to-Cart → Purchase funnel with exact timestamps
License	Free with Kaggle account Open for research	Free with Kaggle account Open for research

Event	Store Equivalent	Count
VIEW	Customer picked up and looked at a product	2,664,312
ADDTOCART	Customer carried the item — strong interest signal	69,332
PURCHASE	Completed buy — strongest signal, reveals basket composition	22,457

3 Entity-Relationship Diagram (Chen Notation)



Rectangles = Entities · Ovals = Attributes · Underlined = Primary Key · Diamonds = Relationships · 1 / N = Cardinality

Entity	Represents	Relationship	Captures
CUSTOMER	The shopper — age group, membership, visit frequency	VISITED	Customer → Zone (movement record)
		TRAVERSED	Zone → Zone (path map, weighted by frequency)
STORE ZONE	Entrance, Tops, Denim, Accessories, Checkout etc.	CO PURCHASED	Article ↔ Article (what gets bought together)
PRODUCT CAT	A product category — Dresses, Footwear, Accessories	PLACED IN	ProductCat → Zone (current placement)
TIME SLOT	When the visit happened — day and hour band	PURCHASED FROM	Customer → Zone (revenue per zone)
ARTICLE	An individual product — specific dress, specific jeans		

4 Business Outcome

The core problem: Retailers place products by instinct. Some zones get 3% of customers, others 94%. Products bought together 70% of the time sit 20 metres apart. Nobody can see the full picture — until now.

Three deliverables on day one:

 Dead Zone Report Every zone ranked by customer reach. Zone 7 gets 3% of visitors. Zone 1 gets 94%. Any product in Zone 7 needs to move — the report says where.	 Top 20 Placement Changes 20 repositioning moves ranked by predicted revenue lift. Each shows: current zone, recommended zone, predicted lift %. Actionable in one afternoon.	 What-If Simulator Drag a product category to a new zone. The ML model instantly shows predicted revenue impact. No more guessing — the number decides.
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Measurable KPIs:

KPI	Target	Measured By
Dead zone elimination	0 zones with under 10% reach	Degree centrality before vs. after
Average basket size increase	+10 to 15%	Avg items per transaction over 30-day window
Co-purchase pair lift	+25% when pairs placed adjacent	CO_BOUGHT edge conversion rate before vs. after
Revenue per sqm improvement	Predicted by ML pre-change	scikit-learn regressor output vs. actual revenue

Why this approach is different:

	Zara / Amazon / Walmart	Our Approach
Technology	RFID + ceiling cameras + proprietary sensor grids	Transaction sequences — every retailer already has this
Cost	£500K – £2M per store installation	Open-source graph tools only — near-zero hardware cost
Privacy	Tracks individuals in real time — growing regulatory risk	Fully anonymised transaction IDs — no personal tracking
Speed to value	12–18 months to install, calibrate, and validate	Insights ready as soon as transaction history is loaded

In one sentence: We turn 31 million H&M purchase records into a store movement map and tell any retailer exactly where every product should live — backed by a predicted revenue number, built on data they already own, with no new hardware required.